

to $(\$/\text{share})^2$. Timing risk (variance) is now the sum of each period's variance over n trading horizons. Taking the square root gives timing risk value in total dollars.

Now suppose that we follow a constant trade rate. That is, the portfolio will be decreasing in a constant manner.

Derivation of the 1/3 factor

As shown above, risk for a specified trade rate is:

$$\mathfrak{R}(\alpha) = \sqrt{X^2 \cdot \frac{1}{250} \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha} \cdot \sigma^2 \cdot \frac{1}{3} \cdot P_0^2} \quad (7.26)$$

The derivation of the 1/3 adjustment factor is as follows:

Let R represent the vector of shares and C represent the covariance matrix scaled for a single period expressed in $(\$/\text{Share})^2$. Then the one-period portfolio risk is:

$$\mathfrak{R}(1) = \sqrt{R'CR} \quad (7.27)$$

For simplicity, we proceed forward using variance (risk squared). This is:

$$\mathfrak{R}^2(1) = R'CR \quad (7.28)$$

The total variance over n periods is an additive function:

$$\mathfrak{R}^2(n) = \underbrace{R'_1 C R_1}_1 + \underbrace{R'_2 C R_2}_2 + \dots + \underbrace{R'_n C R_n}_n = n \cdot R'CR \quad (7.29)$$

For a constant portfolio R , variance scales with the square root of the number of trading periods:

$$\mathfrak{R}(n) = \sqrt{n \cdot R'CR} = \sqrt{n} \cdot \sqrt{R'CR} = \sqrt{n} \cdot \mathfrak{R}(1) \quad (7.30)$$

This is often shown using the time notation as follows:

$$\mathfrak{R}(t) = \sqrt{t} \cdot \mathfrak{R}(1) \quad (7.31)$$

For a portfolio where the share quantities change from period to period, the risk calculation will not simplify as it does above. Risk will need to be computed over all periods. This is:

$$\mathfrak{R}^2(n) = \sqrt{\underbrace{R'_1 C R_1}_1 + \underbrace{R'_2 C R_2}_2 + \dots + \underbrace{R'_n C R_n}_n} \quad (7.32)$$